Indian Institute of Technology (BHU) Varanasi
End-Sem Examination (Odd Sem 2024-25)
MO 201 Materials Science

Time: 3 hours **Full marks: 60**

There are 18 questions in this question paper. Attempt any 12 questions. All questions carry equal marks. If you attempt extra questions, the first 12 answers will be considered, not the best 12 answers. Therefore, choose your questions carefully before you start writing. Non-programmable scientific calculator is allowed.

Q1. Explain what are Shockley and Frank partial dislocations in FCC. [5]

Q2. Explain with diagram what is resolved shear stress, critical resolved shear stress and Schmid factor and what is their importance with respect to plastic deformation. What can be the maximum value of Schmid factor and why? [5]

Q3. (a) What is the difference between slip and twinning? (b) Explain with diagram the formation of a sessile dislocation in an FCC crystal by a dislocation reaction. [2+3=5]

Q4. Define the following with respect to tensile testing (a) Poisson ratio (b) Anelasticity (c) Necking (d) 0.2% offset yield stress (e) modulus of resilience [1x5=5]

Q5. (a) What is ductile to brittle transition in iron? (b) What is Yoffee diagram? [2+3=5]

Q6. Explain the various hardness testing methods Brinell, Rockwell, Vickers and Knoop test. [5]

Q7. (a) Why is carbon black used and not silica as particle reinforcement in vulcanized rubber? (b) Why is steel a suitable reinforcement material for concrete? [2+3=5]

Q8. Derive the equations for elastic modulus of composite in longitudinal and transverse direction for continuous and aligned fiber composites. [2.5x2=5]

Q9. What is fiberglass composite? State its advantages and limitations and typical applications. [5]

Q10. Draw the iron-carbon phase diagram showing all the phases and invariant reactions. [5]

Q11. Explain the principle of zone refining from phase diagram. [5]

Q12. Explain the free electron theory and on the basis of the same, explain why solids conduct electricity. [5]

Q13. Explain (a) intrinsic semiconductor (b) p-type extrinsic semiconductor (c) n-type extrinsic semiconductor. [1+2+2 =5]

Q14. Explain (a) variation of conductivity with temperature for an extrinsic semiconductor (b) forward and reverse bias in junction diode. [3+2 =5]

Q15. (a) Explain ferromagnetism in Heusler alloys although Mn is not ferromagnetic in elemental form. (b) Explain what happens to the magnetization of $NiFe_2O_4$ upon addition of Zn. (c) What is ferrimagnetism? [2+2+1=5]

Q16. (a) Define the various polarization mechanisms. (b) Explain why $BaTiO_3$ shows ferroelectricity. [2+3=5]

Q17. What are (a) cermets and (b) hybrid composites? (c) What is the importance of critical fiber length? [2+2 +1=5]

Q18. Explain the types of structural composites. [5]

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